



**RPOW**  
EARLY BETA  
rpow.net

4758

**A CYPHERPUNK'S MANIFESTO**  
ERIC HUGHES • 1993

# REUSABLE PROOF OF WORK

004 / 15 AUG 2004 / HAL FINNEY / RARE

## BUT THEY ARE REUSABLE.

In the dark workshop, Hal Finney holds up a bright new token. A stamped envelope waits in the input tray, while an older token sits inside the machine. The warm light falls across his face and the secure hardware. This is an imagined way to show a change in what proof of work could do: a proof used to make something that could be passed on.

On 15 August 2004, Finney invited the cypherpunks mailing list to test his new server at rpow.net. It accepted a Hashcash proof-of-work token and returned an RSA-signed token he called a **Reusable Proof of Work**, or RPOW. A recipient could later exchange that RPOW for a fresh one of equal value. Each individual token could be spent only once, yet the chain of exchanges let the value move from person to person. As Finney put it, unlike Hashcash, **“they are reusable.”**

The promise depended on trust in the machine doing the exchange. Finney ran the server on an IBM 4758 secure cryptographic coprocessor. Its signed attestation was designed to let a sufficiently motivated user check the software configuration against the published source code. That was the safeguard behind his claim that the server would issue a new token only when it received an equal-value proof. RPOW was an experiment with a trusted server, not Bitcoin or a decentralised blockchain.

Finney called the system an early beta and warned users that a server reload might invalidate tokens already made. The empty output tray in our scene catches that uncertainty alongside the excitement. In the foreground, Eric Hughes's earlier manifesto and the Hashcash envelope connect cards 002 and 003 to this one. Their presence is HISTROVE's visual storytelling, not a claim that these objects sat on Finney's actual desk.

### HIDDEN IN THE ART

#### 01 THE RETURNING ENVELOPE

The small cream envelope in the left input tray bears the blue digital approval mark seen on card 003, Hashcash. It stands for the proof Finney's server could accept, not a physical letter documented in the RPOW experiment.

#### 02 THE PRINTED MANIFESTO

The foreground page is headed A Cypherpunk's Manifesto - Eric Hughes · 1993, linking back to card 002 and the mailing-list culture in which RPOW was announced. It is a copy of Hughes's text, not something Hal wrote; the card frame partly hides it, while the full book art reveals it.

#### 03 THE 4758 PLATE

The number on the machine refers to the IBM 4758 secure coprocessor that hosted the server. The large exchange apparatus around it is an artistic metaphor rather than a reconstruction of the real hardware.

#### 04 THE TWO TOKENS

One token remains inside the window and a new gold one shines in Hal's hand. Their shared black recycling emblem, supplied for this artwork, helps show single-use proofs giving rise to replaceable RPOWs. No physical gold RPOW coins were minted in this scene.

#### 05 THE BETA SCREEN

The comic-style RPOW mark, EARLY BETA and rpow.net echo Finney's public invitation and the user-supplied logo reference. The logo's exact historical provenance has not been established, so the screen is not presented as a verified capture of his interface.

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### SOURCE & ART NOTE

Hal Finney, *RPOW - Reusable Proofs of Work*, 15 August 2004; Eric Hughes, *A Cypherpunk's Manifesto*, 9 March 1993. The workshop, exchange apparatus, physical tokens and manifesto page are illustrations. The fine text printed beneath the manifesto heading is decorative and should not be treated as a reproduction of Hughes's essay. The source describes RPOW as an early beta on a secure server and warns that a reload could invalidate tokens.